Homework4

Question 11.1

1. **Stepwise Regression (Linear Regression):**

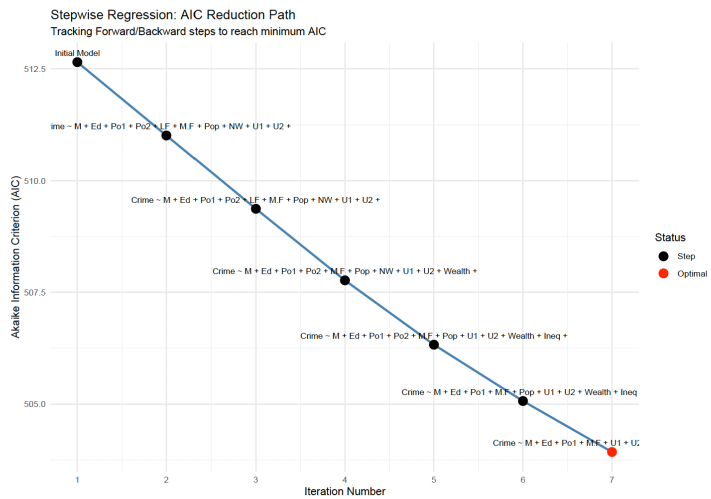
   Implementation: The implementation uses a bidirectional approach (both forward and backward) to minimize the Akaike information criterion (AIC).

   - **Initial State:** The implementation starts with a full linear model (lm(Crime ~ ., data = data)) containing all 15 predictor variables.
   - **Iterative Process:** The step() function evaluates adding or removing variables one ata time. The script captures this "path" by parsing the step_log to extract AIC values and the specific variable changes at each iteration.
   - **Optimization:** The process stops when no further additions or removals can lower the AIC, identifying the most statistically significant subset of predictors.

   Results and Analysis:

   - **Model Fit:** The final model achieved an Adjusted R-squared of 0.7444, explaining approximately 74.4% of the variance in crime rates.
   - **Residuals:** The Residual Standard Error is 195.5 on 38 degrees of freedom.
   - **Significant Predictors:** The summary output highlights several key variables with high significance ($p < 0.01$) such as M (percentage of males), Ed (mean years of schooling), Ineq (income inequality), and Prob (probability of imprisonment).

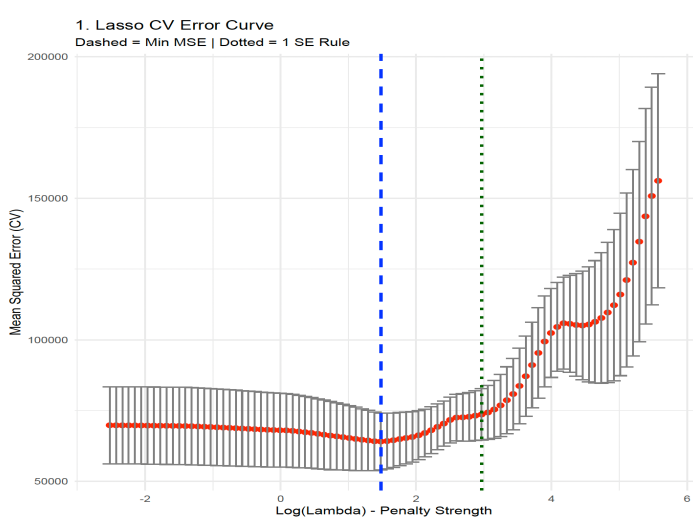
Predictor (Variable)	Estimate (Coefficient)	Std. Error	t value	Pr(>|t|)	Significance
(Intercept)	-6426.1	1194.61	-5.379	0.00000404	***
M (Percentage of Males)	93.32	33.5	2.786	0.00828	**
Ed (Mean Years of Schooling)	180.12	52.75	3.414	0.00153	**
Po1 (Per Capita Expenditure on Police)	102.65	15.52	6.613	8.26E-08	***
M.F (Number of Males per 1000 Females)	22.34	13.6	1.642	0.10874	
U1 (Unemployment Rate, Urban Males 14–24)	-6086.63	3339.27	-1.823	0.07622	.
U2 (Unemployment Rate, Urban Males 35–39)	187.35	72.48	2.585	0.01371	*
Ineq (Income Inequality)	61.33	13.96	4.394	0.0000863	***
Prob (Probability of Imprisonment)	-3796.03	1490.65	-2.547	0.01505	*



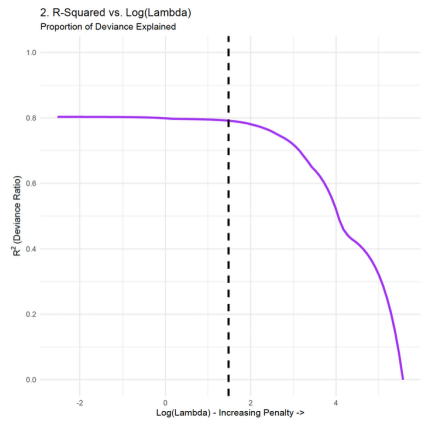
2. Lasso Regression ($\propto = 1$):

Implementation: Lasso (Least Absolute Shrinkage and Selection Operator) is implemented via the glmnet package, which adds an L1 penalty to the loss function.

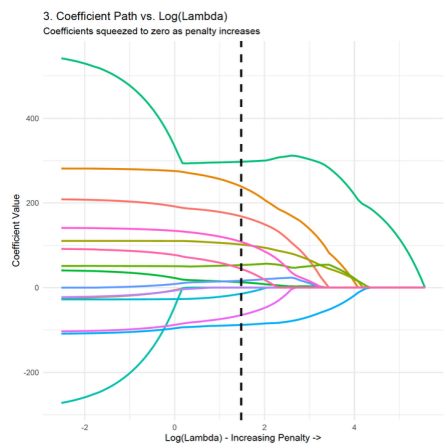
- **Standardization:** Predictors are scaled (scale(x)) before fitting to ensure the penalty is applied uniformly across different units.
- **Tuning:** 5-fold cross validation (cv.glmnet) is used to find the optimal λ $(penalty\ strength)$.
- **Visualization:** The script generates a four-panel diagnostic dashboard:
  - **CV Error curve:** Shows the mean squared error (MSE) across different log(λ) values.



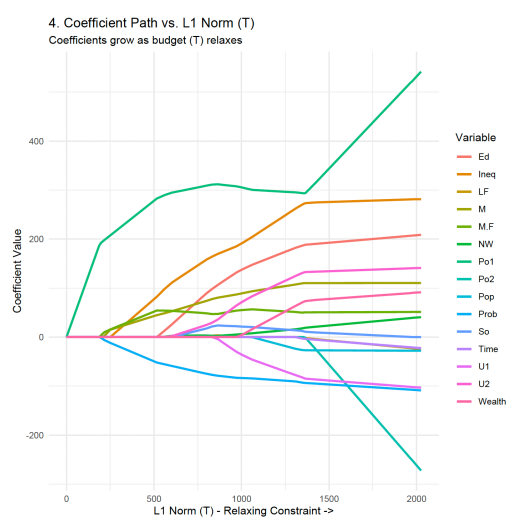
- **R-Squared curve:** Tracks the proportion of deviation explained as the penalty increases.



- **Coefficient Path (Lambda):** Shows coefficients being "squeeze" to exactly zero.



- **Coefficient Path (L1 Norm):** Visualizes variable entry as the "budget" (T) for coefficients relaxes.



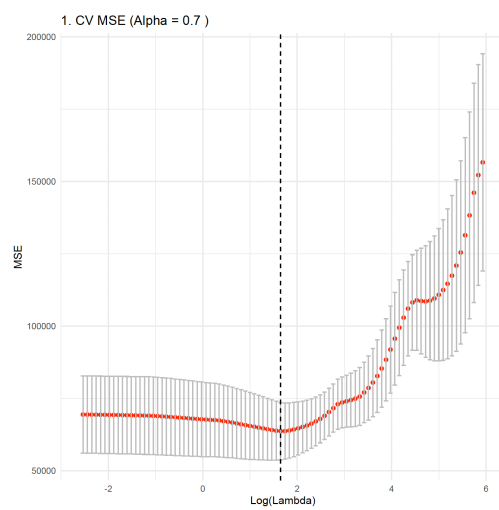
Results and Analysis:

- **Variable Sparsity:** Lasso successfully performs variable selection by zeroing out less impactful predictors, such as So, Time, and Wealth, as seen in the coefficient paths.
- **Optimal Penalty:** The "Dashed" line in the diagnostic plots indicated "lambda.min", while the "Dotted" line shows the simpler "lambda.lse" model.

3. **Elastic Net Regression ($\propto = \mathbf{0.7}$):**

Implementation: Elastic Net combines L1 (Lasso) and L2 (Ridge) penalties. The implementation includes an automated hyperparameter search.

- **Alpha Search:** The code loops $\propto$ values from 0.0 to 1.0 (step 0.1) and selects the one with the lowest CV MSE.
- **Selection:** The search identified $\propto = \mathbf{0.7}$ as the optimal mixing parameter.
- **Final Model:** A final cv.glmnet is performed at $\propto = \mathbf{0.7}$ to select the best λ.
    - **CV Error curve:** Shows the mean squared error (MSE) across different log(λ) values.



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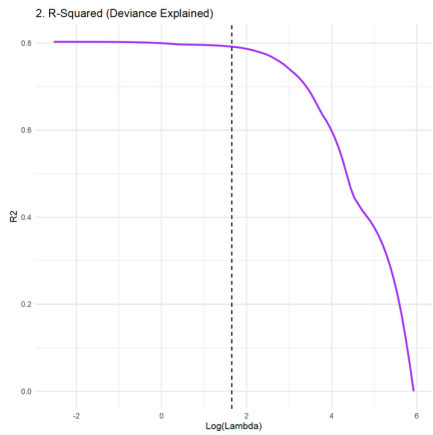

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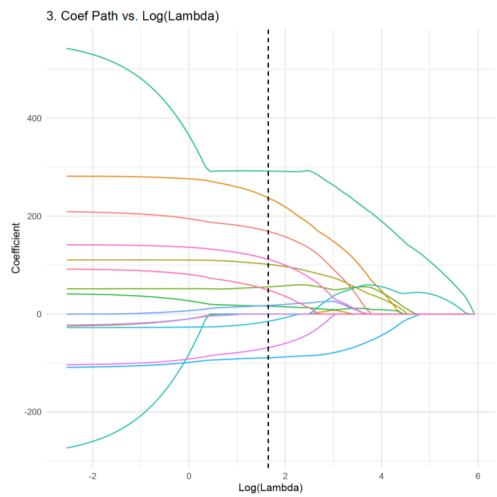

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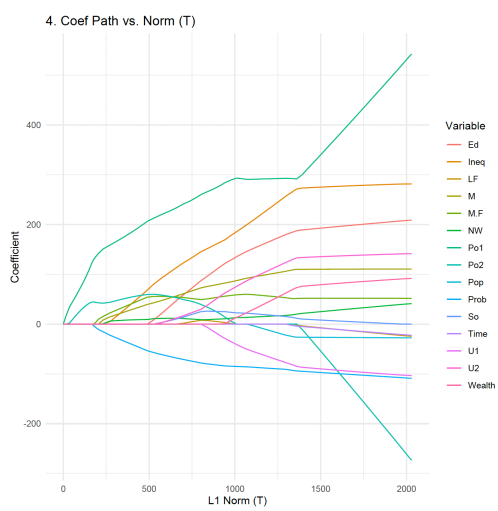

Results and Analysis:

- **Hybrid Behavior:** With $\propto = \mathbf{0.7}$**,** the model behaves more like Lasso than Ridge but retains enough influence to handle potential multicollinearity between variables like Po1 and Po2.
- **Predictive Power:** The R-squared plot for the Elastic Net shows a high peak deviance ratio (approx. 0.7-0.8) before the penalty becomes too severe.
- **MSE Performance:** The CV MSE curve for $\propto = \mathbf{0.7}$ shows a clear "U-shape", identifying a precise λ that balances bias and variance better than a standard linear model.

Predictor (Variable)	Coefficient (at lambda.min)	Status
(Intercept)	905.08511	Included
M (Percentage of Males)	101.3916	Included
So (Southern State)	17.44893	Included
Ed (Mean Years of Schooling)	168.65087	Included
Po1 (Expenditure on Police '60)	291.89829	Included
Po2 (Expenditure on Police '59)	0	Dropped (Zeroed)
LF (Labour Force Participation)	0	Dropped (Zeroed)
M.F (Number of Males per 1000 Females)	55.42468	Included
Pop (State Population)	-14.82519	Included
NW (Number of Non-whites)	16.07239	Included
U1 (Unemployment Rate, Urban Males 14–24)	-68.63263	Included
U2 (Unemployment Rate, Urban Males 35–39)	111.58679	Included
Wealth (Wealth Index)	48.87439	Included
Ineq (Income Inequality)	237.24046	Included
Prob (Probability of Imprisonment)	-89.34071	Included
Time (Average Time in Prison)	0	Dropped (Zeroed)

4. Detailed Comparison:

Feature	Stepwise Regression	Lasso (α=1)	Elastic Net (α=0.7)
Methodology	Iterative addition/removal based on AIC.	Continuous shrinkage and automatic variable selection.	Hybrid shrinkage; groups correlated variables together.
Variable Selection	Results in a hard "in or out" subset.	Can shrink coefficients to exactly zero.	Can shrink to zero but is less aggressive than Lasso.
Stability	High variance; small changes in data can change the subset.	Can be unstable if variables are highly correlated.	Most stable; handles multicollinearity via the Ridge component.
Key Result	Adjusted R2=0.7444.	Sparse model; removes noise.	Optimal balance found at α=0.7.

The **stepwise** model provides the most interpretable list of predictions but risks overfitting. **Lasso** is effective for automated feature selection but may arbitrarily pick one variable from a correlated group. **Elastic Net** ($\propto = \mathbf{0.7}$) is the superior implementation for this dataset as it provides the predictive benefits of Lasso while using a hybrid penalty to maintain model stability across the 15 crime-related predictors, effectively handling potential multicollinearity between related variables like Po1 and Po2. It also performed variable selection by shrinking the coefficients of Po2, LF and Time to exactly zero. It also showed variables with largest positive impact are Po1 (291.90), Ineq(237.24), and Ed(168.65), indicating they are strong positive predictors of crime in this model.